वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)

यूनियन बैंक ऑफ इंडिया Union Bank of India

PRESENTS

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Ai - CSPARC

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration

SOMAIYA VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
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PROBLEM STATEMENT & TEAM DETAILS

PatternX

TEAM NAME (AS SUBMITTED ON PORTAL)		**NeuralNova**	
PROBLEM STATEMENT TITLE		**AI Powered Explainable Graph-Based Fraud Detection System** Making financial fraud visible, explainable and preventable	
	NAME	**GENDER**	**AREA OF EXPERTISE**
Member 1	Mahak Bajpai	Female	Team Leader | Creative Direction and Strategy
Member 2	Srishti Singh	Female	Frontend | UI/UX
Member 3	Medha Saxena	Female	AI/ML Integration
Member 4	Sachin Yadav	Male	Mern Stack Developer (Backend | DB)

PatternX- AI Powered Graph based fraud detection system

PatternX

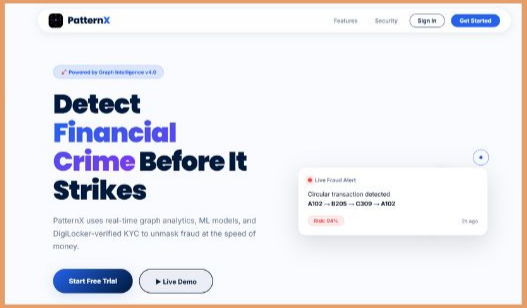
Problem Faced

India loses 1000/- + crore annually due to complex fraud patterns undetected by traditional system. This is because current banking systems analyze transactions individually, making it difficult to detect patterns like circular transactions, layering, structuring, etc.

Unlike traditional systems, PatternX not only detects fraud but also explains the reason behind it and suggests actionable steps

Our Solution - PatternX

- A **graph-based AI fraud detection system** that transforms transaction data into a **connected network of accounts and transactions.**
- By representing **accounts as nodes** and **transactions as edges**, our system enables **real-time tracking of fund movement**.
- PatternX **identifies hidden patterns** by analyzing **relationships between transactions**, enabling faster and more accurate detection.
- The system detects key fraud types including **circular transactions, layering, structuring, dormant account misuse, and high-frequency activity.**



Transactions → **Graph Network** → **Fraud Detection** → **Risk Score** → **Explanation**

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

PatternX

Risk Scoring Engine

PatternX assigns a dynamic risk score (0–100) based on multiple behavioral and structural features.

Features used:

- Transaction frequency
- Amount patterns
- Graph patterns (cycles, chains)
- Profile mismatch
- Account history
- Account Age
- Time Pattern

Risk Score = Σ (Pattern Score + Behavior Score + Profile Score)

Fraud v/s Legit Differentiation

Context-Aware Decision System

Similar patterns can appear in both fraud and legitimate activity. PatternX uses behavioral context to differentiate between them. PatternX :-

- **Matches with historical behavior**
- **Checks transaction timing patterns**
- **Analyzes network structure**

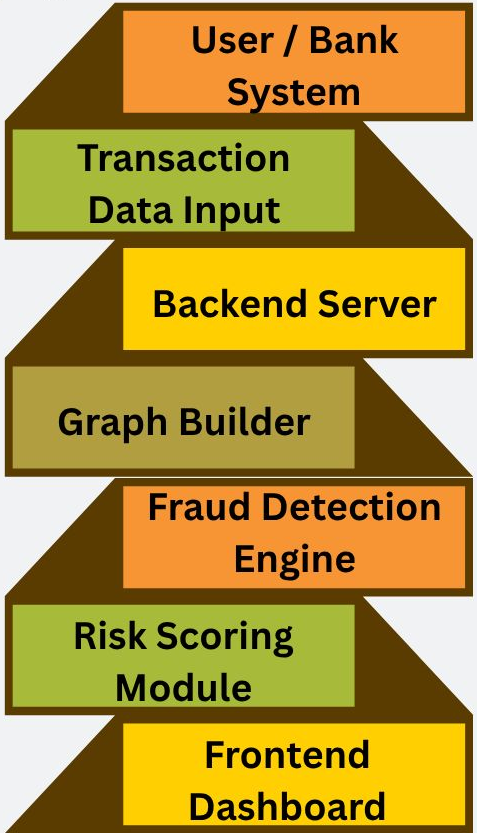
Detection is based not only on pattern, but on pattern + context to prevent flagging any legit transaction

Fraud Type Handling

Circular Transactions	• Detects cycles in graph • Flags round-tripping
Layering	• Identifies long transaction chains • Detects money hiding patterns
Structuring	• Detects multiple small transfers • Within short time window
Dormant Acc Misuse	• Detects sudden activity spikes • After inactivity
High Frequency Activity	• Detects burst transactions • Time-based anomaly

Instead of binary classification, our system provides graded risk levels — reducing false positives.

System Architecture



TECHNICAL APPROACH

PatternX

Methodology

1. Capture transaction data
2. Convert into graph structure
3. Detect fraud patterns
4. Calculate risk score
5. Generate explanation
6. Display results on dashboard

Tech Stack

Frontend	React.js
Backend	Node.js
Database	MongoDB
Graph Visualization	Cytoscape.js
Logic	Graph based anomaly detection + behaviour learning models

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PatternX

FEASIBILITY & VIABILITY

Technical Feasibilty

PatternX is built using scalable and widely adopted technologies such as React, Node.js and graph-based processing, making it feasible for real-world deployment.

- Compatible with current banking systems (API-based integration)
- No additional hardware required
- Can scale with increasing data
- Uses existing banking transaction data

Potential Challenges

- Handling large-scale transaction data
- Risk of false positives
- Real-time processing requirements
- Data privacy and security concerns

Solutions

→ Use efficient graph algorithms + scalable backend

→ Risk scoring + behavioral analysis + profile matching

→ Stream processing + optimized logic

→ Encryption + secure APIs

Our system is designed to balance accuracy, scalability, and security. PatternX is not just a concept — it is a practical and deployable solution for modern banking systems.

IMPACT & BENEFITS

Real World Impact

1. **Enables real-time fraud detection**
2. **Improves transparency in transaction monitoring**
3. **Reduces financial losses for banks**
4. **Speeds up fraud investigation process**

Unlike existing systems , PatternX detects fraud networks, not just transactions.

Benefits

🏦 Banks
- Reduced fraud losses
- Faster investigation
- Better compliance

👥 Customers
- Increased trust
- Safer transactions
- Protection from fraud

🌐 Economy / Society
- Reduced financial crime
- Stronger digital banking ecosystem
- Improved financial security

PatternX makes financial systems smarter, safer, and more trustworthy.

BUSINESS MODEL

PatternX

Revenue Model

PatternX will be offered as a SaaS (Software-as-a-Service) solution for financial institutions.

Revenue Streams

- Subscription-based pricing (monthly/yearly)
- API integration charges
- Premium analytics & reporting features

Flexible pricing based on transaction volume and scale of deployment. From detection to decision — all in one system

Target Market

- Banks (Public & Private)
- NBFCs
- Digital payment platforms (UPI apps, wallets)

Value Proposition

Real-time, explainable fraud detection with reduced false positives and actionable insights.

🌍 Market Potential:

- Growing digital transactions
- Increasing fraud cases globally
- High demand for intelligent fraud systems

🚀 Scalability:

- Cloud-based → easy to scale
- Works across multiple banks
- Can integrate with UPI / fintech ecosystems

🛠️ Go-To-Market Strategy:

- Partner with banks & fintech companies
- Pilot testing with small datasets
- Gradual rollout

PatternX is not just a solution — it is a scalable business opportunity in the growing fintech security market.

RESEARCH & REFERENCES

References

RBI Annual Report on Banking Frauds

Graph Based Fraud Detection Research Paper

Anti Money Laundering Research Paper

AI in financial security

Our solution is inspired by modern research in graph-based fraud detection and AI-driven financial security systems.

Here is one page overview of PatternX explaining entire concept in detail

PatternX One page summary